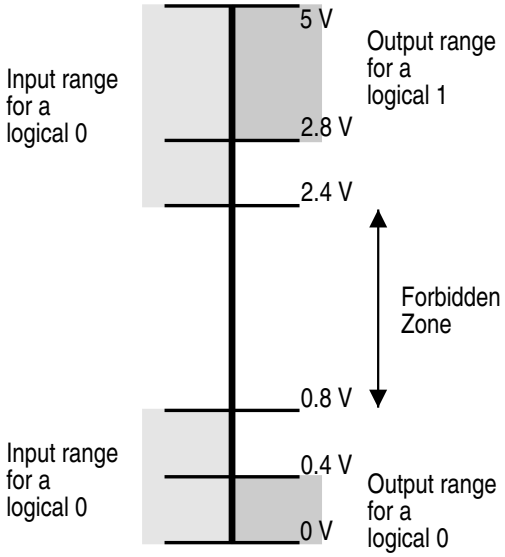
Systems Architecture

Information, Data or Program

Two State Device



Interpreting States

- Either 1 or 0, no intermediate
  - ON or OFF
  - High or Low
  - True or False
- Unipolar Device:
  High Voltage indicates a logical 1 / True / On
  Low Voltage indicates a logical 0 / False / Off
- Bipolar Device:
  Positive Voltage indicates a logical 1
  Negative Voltage indicates a logical 0

Information from Bits

- Each Two-State Device represents a *Bi*nary Dig*it*
- Group *BIT*s together to form larger values:

Bits		*Name*	*Comment*	*Values*
2^0	1	Bit	Binary Digit	2
2^2	4	Nibble	Half a byte	16
2^3	8	Byte	two nibbles	256
2^4	16	Word	two bytes	65,536
2^5	32	Long Word	two words	4,294,967,296

- Values can be interpreted as information:

Numbers	Signed/Unsigned Integer/Floating-Point
Text	Characters (ASCII / Unicode)
Graphics	Colour of a picture element (pixel)
Program	Program instructions (Byte/Native code)

Numbers

Decimal Numbers

	10^3	10^2	10^1	10^0	
	1000	100	10	1	
$\times$	2	3	8	7	
$\sum$	2000	300	80	7	$= 2387_{10}$

Binary Numbers

	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
	128	64	32	16	8	4	2	1	
$\times$	0	1	1	0	0	0	1	1	
$\sum$	0	64	32	0	0	0	2	1	$= 99_{10}$

Characters

- Characters are represented as numbers
- **ASCII**
  (American Standard Code for Information Interchange)

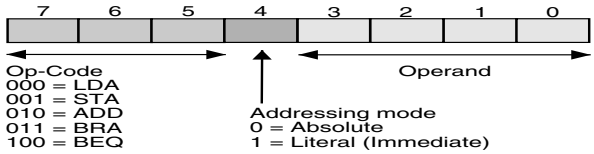
  One Byte per character, 127 “characters” defined including Upper- and Lower-case (American) alphabet numbers, punctuation, and control codes
- **Unicode** — Unified Code
  International Standard not just American

  Two Bytes per character, one for character set and one for character in the set.

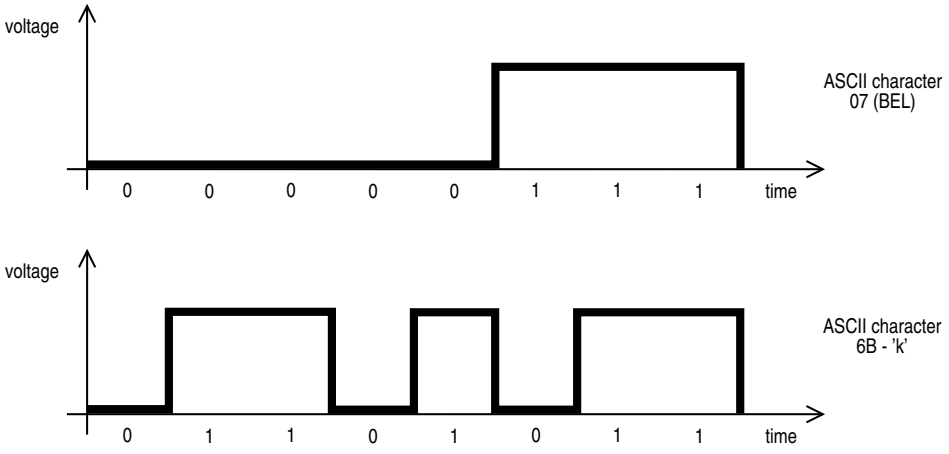
  All major character sets included (including ASCII) Still under development.

Program Instructions

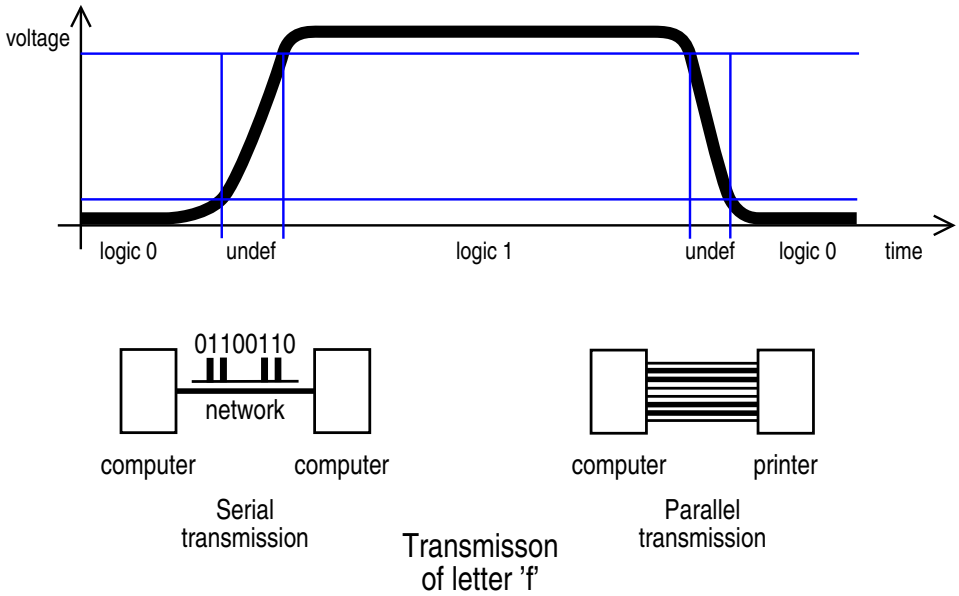
- **Compiler**
  Converts High level language (such as C) into Assembler instructions
- **Assembler**
  Converts Assembler Language into machine instructions.
- Central Processing Unit (CPU)
  Machine instructions tell CPU how to move data around the system.
- Example of a simple 8-bit machine instruction



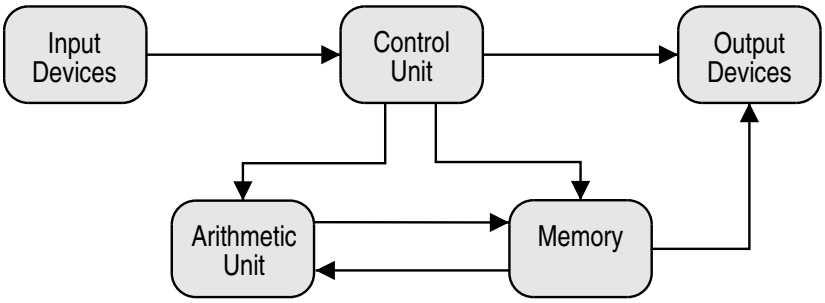
Data



Sending Data



Computers



Data passes between components

Data is both Information and Program Instructions

The Internet

- **Network** / **Intranet**
  Collection of Independent Computers connected together to allow data to pass between them
- **Internet**
  Collection of connected Independent Networks
- **Services**
  eMail / News / World Wide Web / File Transfer

  Work by passing data between computers on different networks